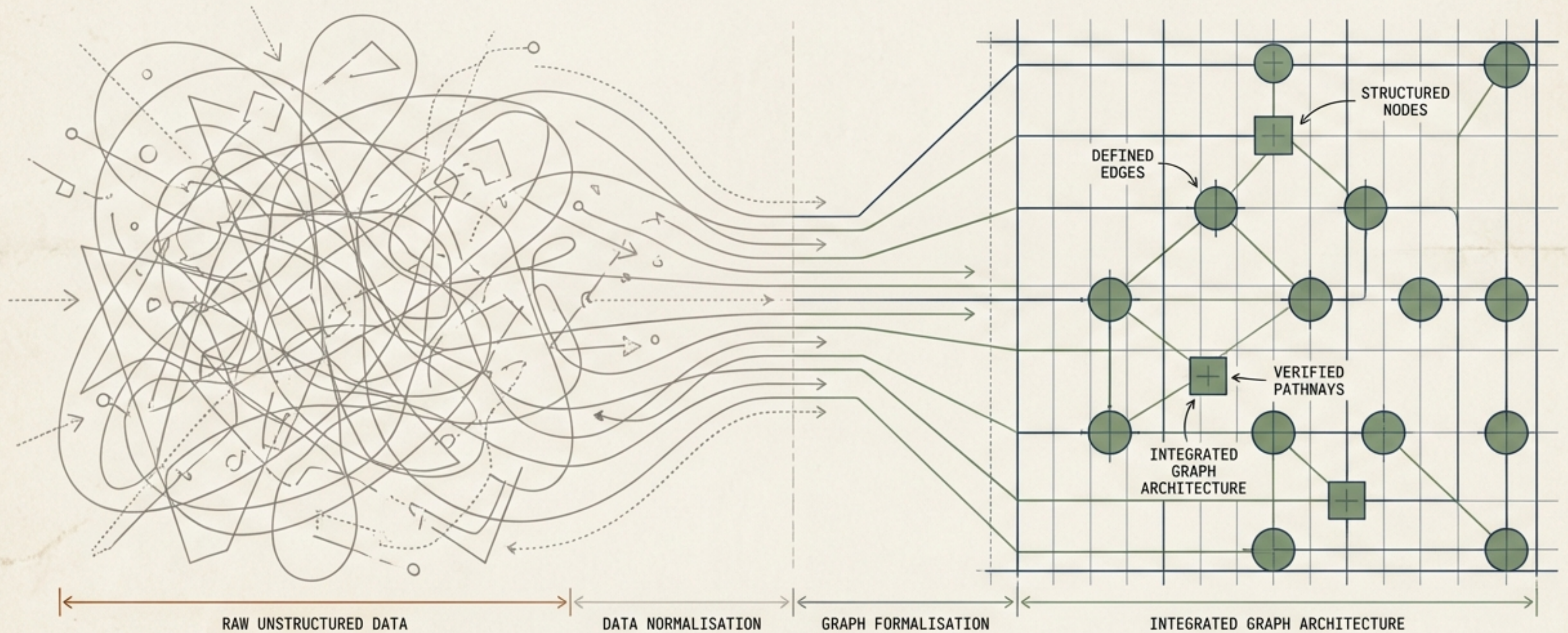
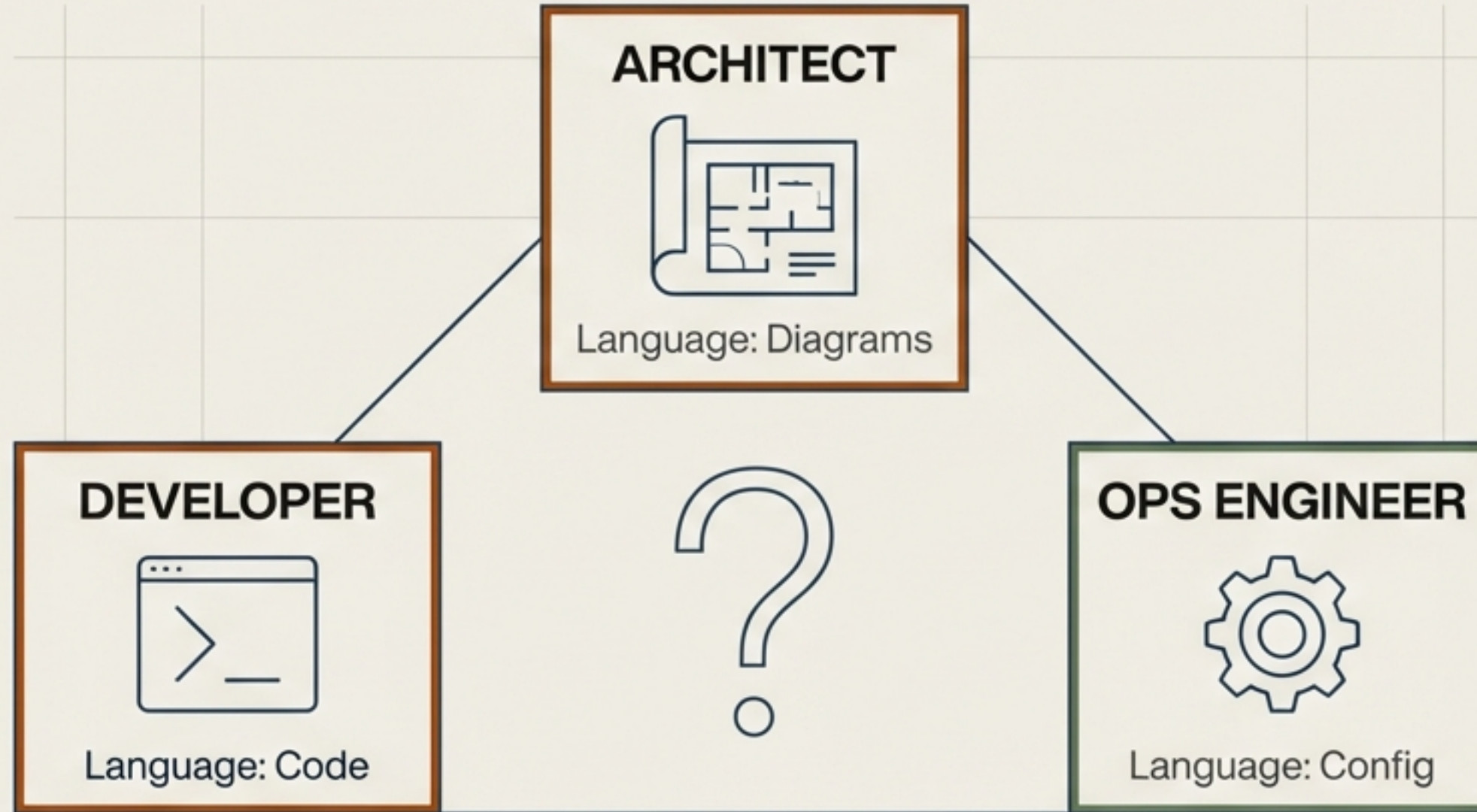


Compatibility Through Connectivity

The Evolution of a Graph-First Architecture for System Verification



The Disconnect Between Design and Reality

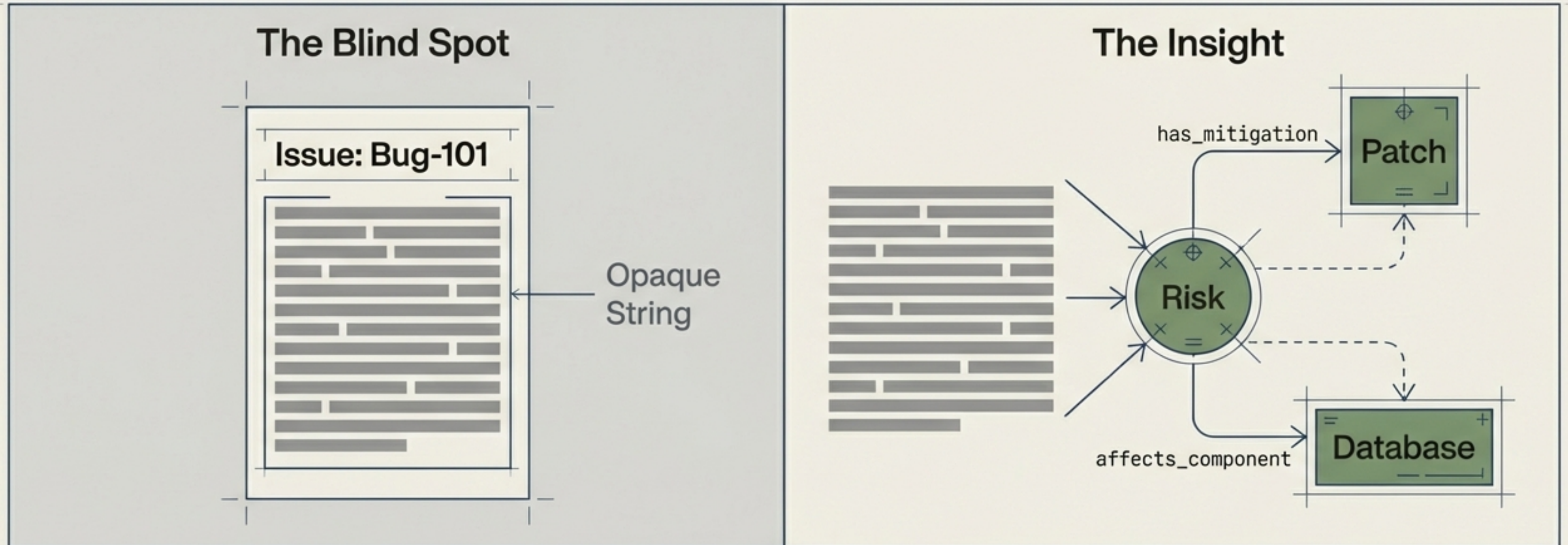


The Critical Question: Does the system work the way the architect **thinks it works?**

~~Goal: Verify Correctness~~ → **Goal: Verify Agreement**

Text is a Graph

Turning Point 1: Making the Opaque Transparent

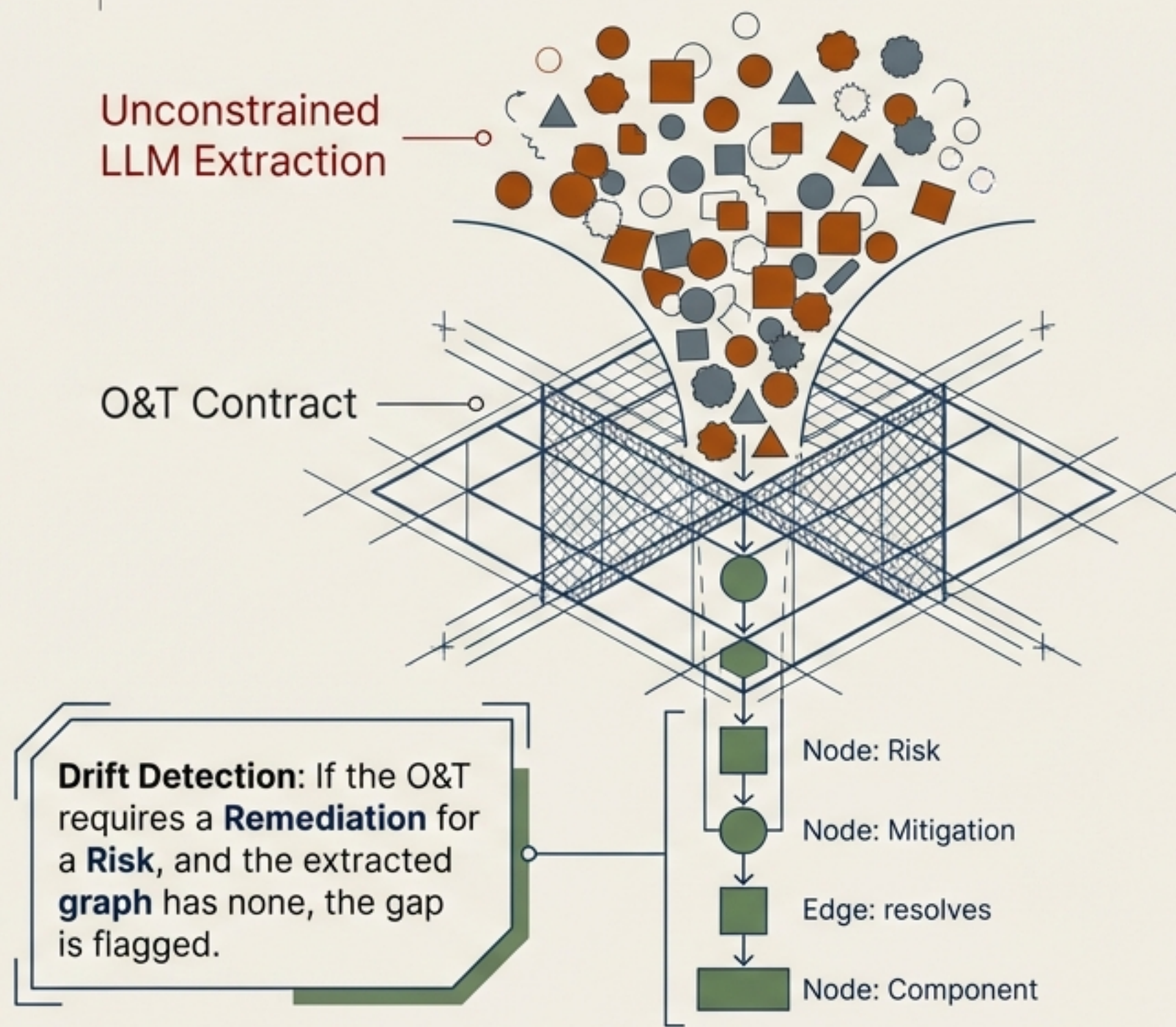


The Voice Memo Insight: If meaning comes from connectivity, we must traverse the text boundary.

Preventing Sprawl with an O&T Contract

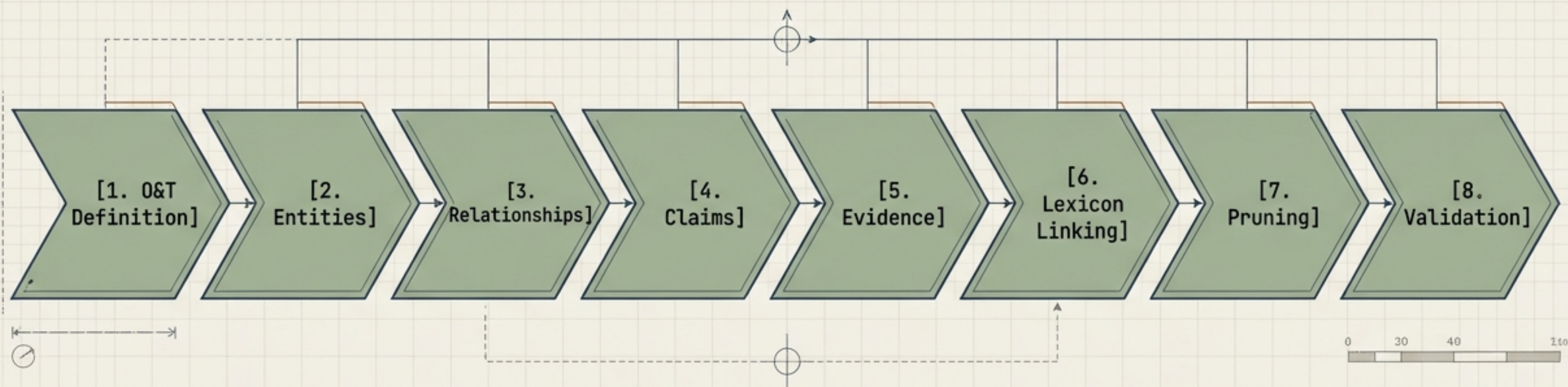
The O&T First Principle

- 1. Define Ontology (Node Types).
- 2. Define Taxonomy (Edge Types).
- 3. Define Constraints.
- 4. Extract *into* the contract.



Precision Through Multi-Pass Extraction

Separation of Concerns across 8 Distinct Agents



The Hurdle: Single-pass prompts fail at complexity. Specialised agents build the graph incrementally.

Measuring Meaning, Not Characters

The Difference Between Text Diff and Graph Diff

Standard Text Diff (Noise)

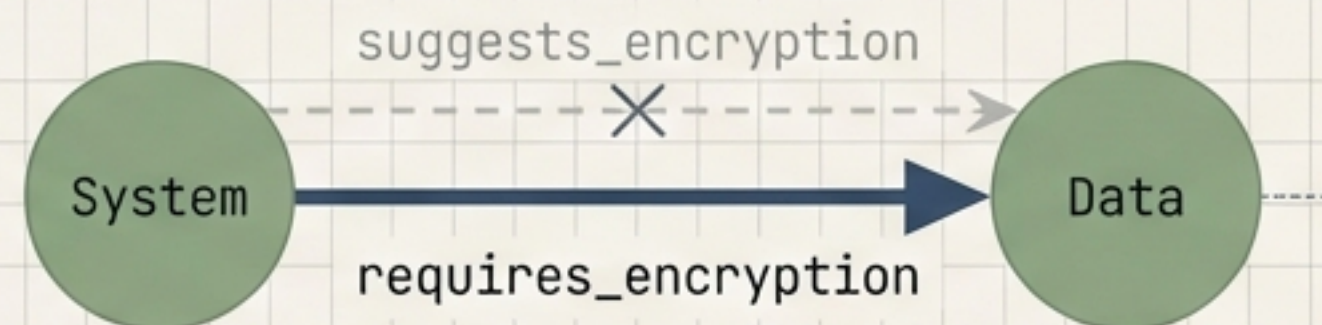
The system **should** encrypt data.

×

The system **must** encrypt data.

Character Change Detected

Semantic Graph Diff (Signal)



Relationship Logic Change Detected

Use Cases

Hallucination Detection
(Claims without evidence).

Translation Validation
(Modal verbs weakening).

Quote Verification
(Dropped negations).

Tests Must Target Code, Not Text

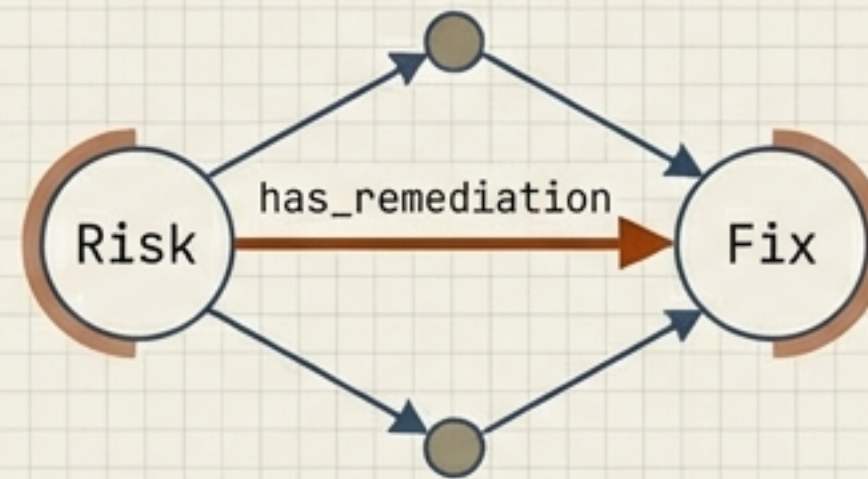
Solving the Brittleness of Gherkin

Natural Language

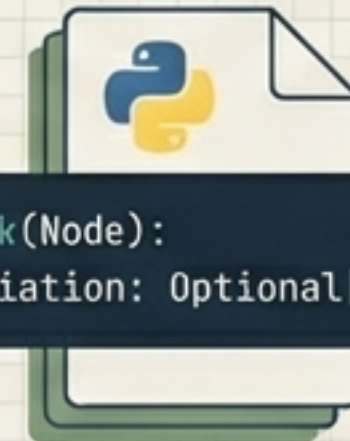


“The risk needs a fix”

Semantic Graph



Code Representation



```
class Risk(Node):  
    remediation: Optional[Remediation]
```

Test Logic:

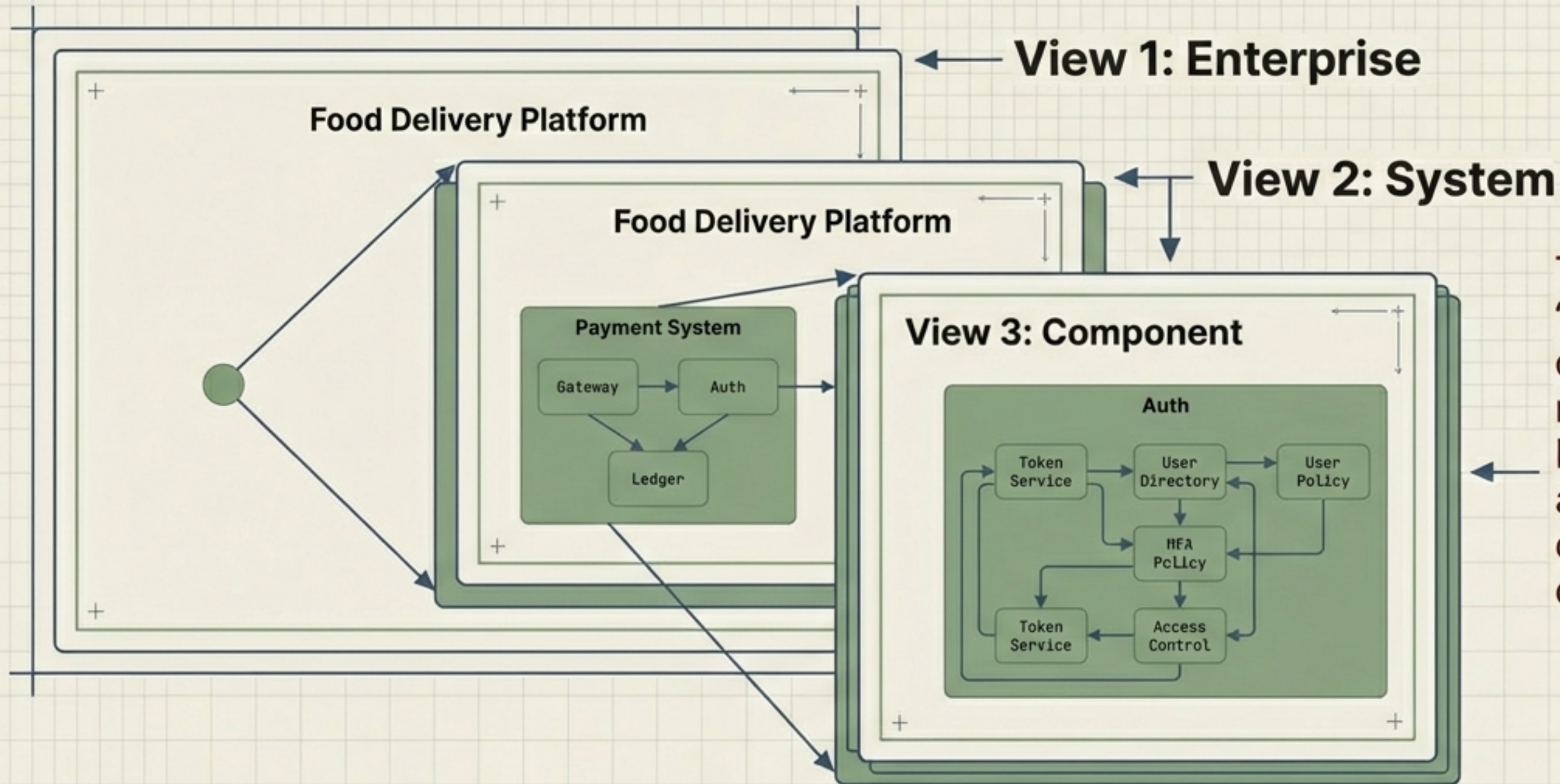
```
assert risk.remediation is not None
```

Benefit:

This test passes regardless of synonyms (fix, patch, mitigation) because the structure is identical.

Reality is Fractal, Not Fixed

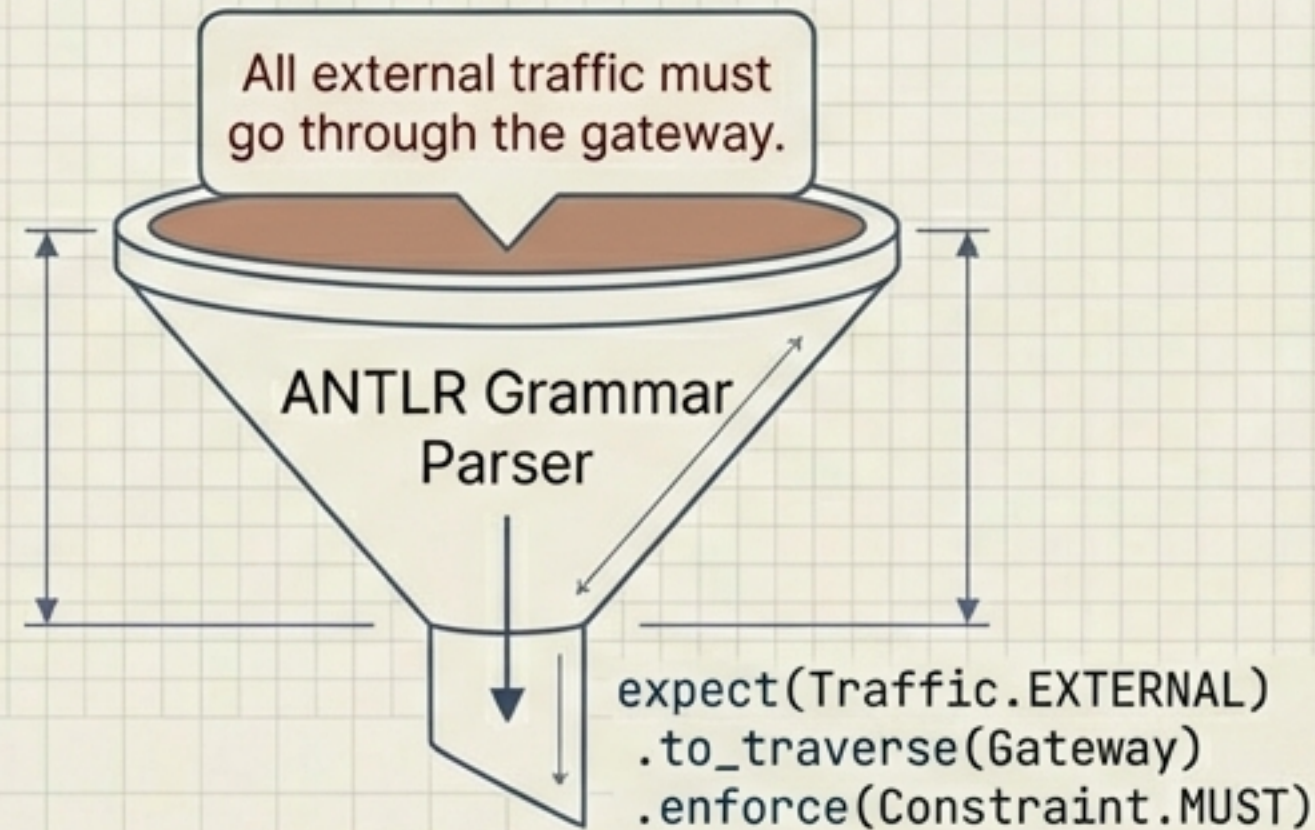
The Critique of the C4 Model



The Insight:
"Level" is a property
of traversal, not the
node. An object can
be a component or
a system depending
on the observer's
distance.

Formalising the Architect's Voice

Parsing Natural Language into Executable Logic



The Vocabulary of Intent

Subjects

all, any,
no,
every

Relationships

uses,
contains,
talks_to

Constraints

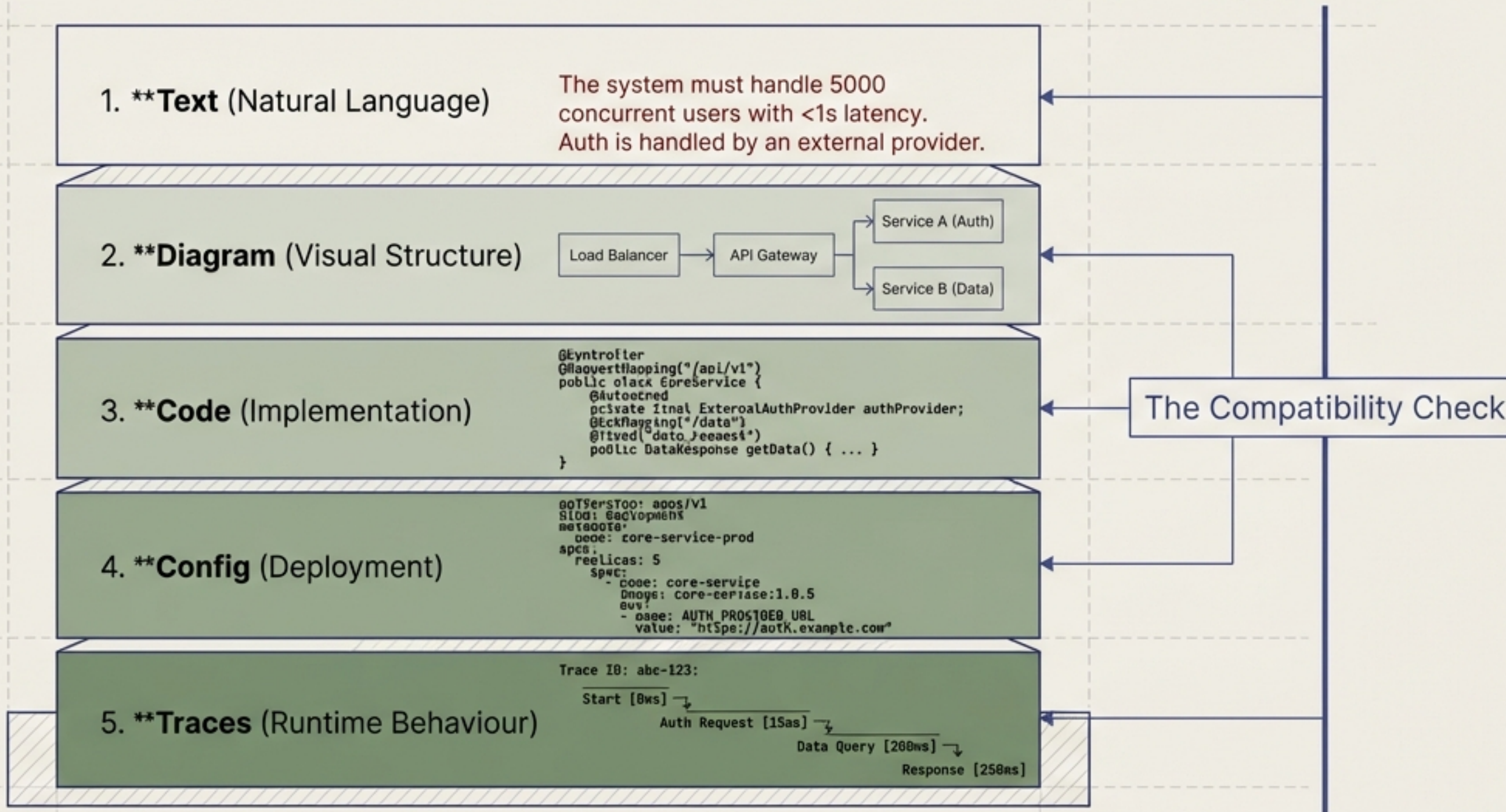
must,
should,
cannot

Workflow

Voice → Parse → Fluent API → Test Code

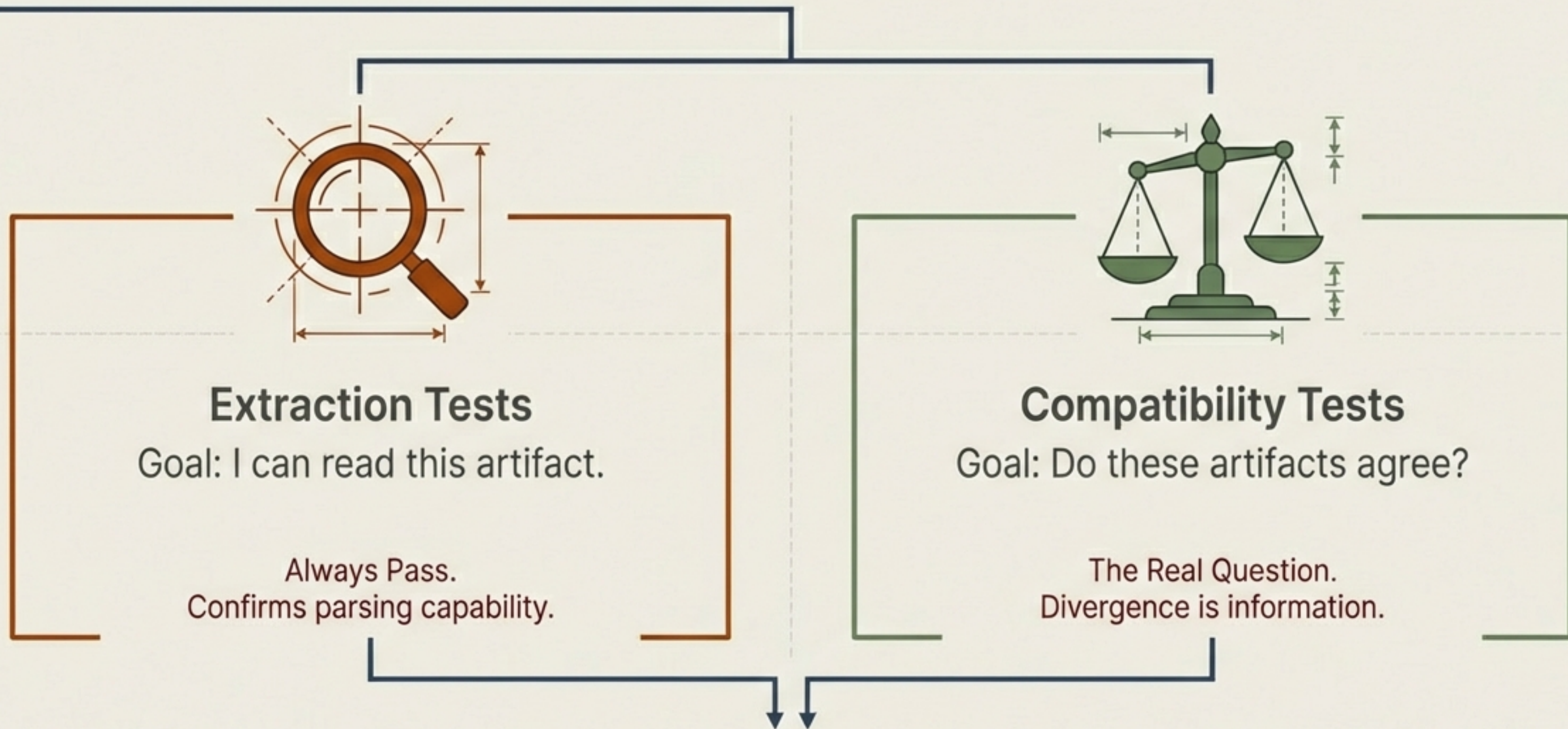
The Grand Unification: Five Layers

Ensuring Consistency Across Multiple System Representations



We have five different representations of the same system. The goal is to verify that they all tell the same story.

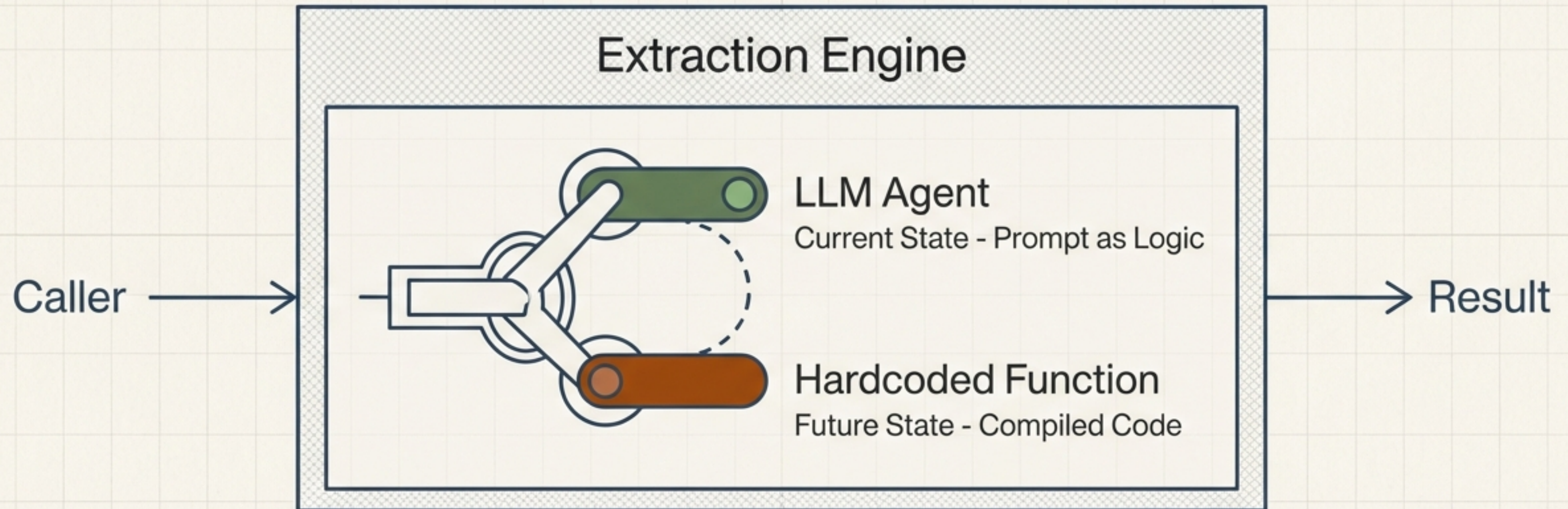
Two Types of Verification



Separating the capability to 'see' (Extraction) from the judgment of 'alignment' (Compatibility).

The Prompt is the Specification

Solving the Bootstrap Problem



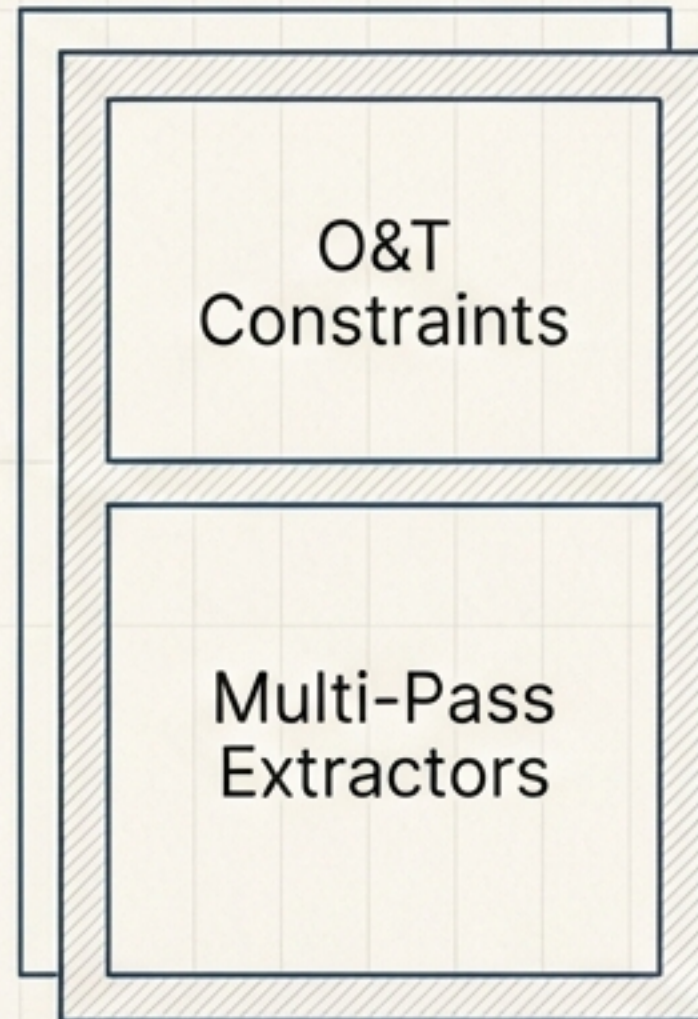
The 'Caller' does not know the implementation. We use LLMs today to validate the architecture, then replace them with code for performance.

The Complete Architecture

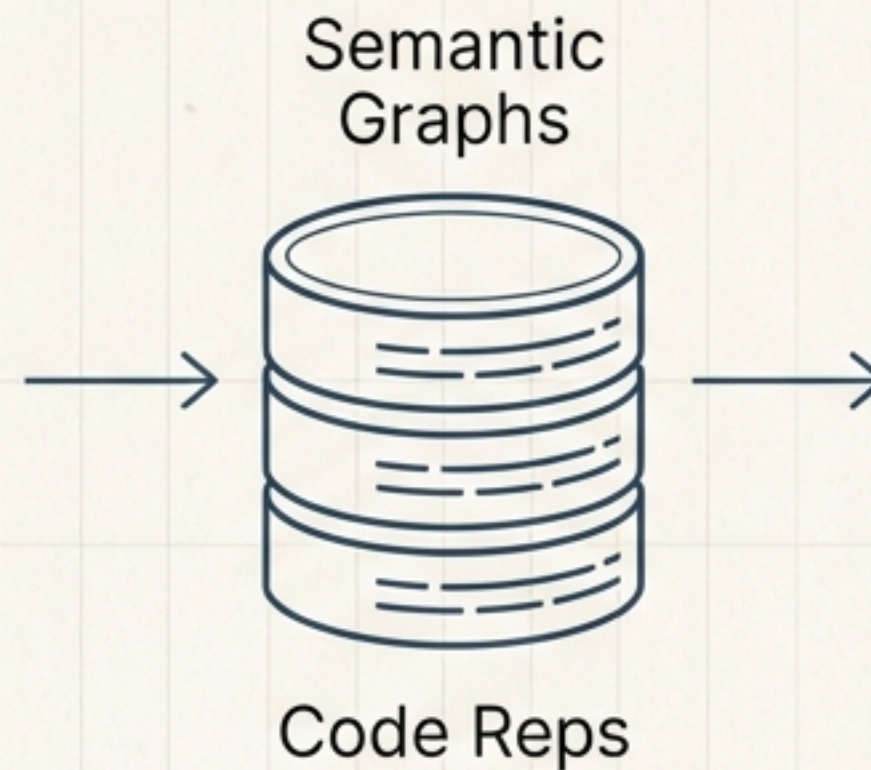
Sources



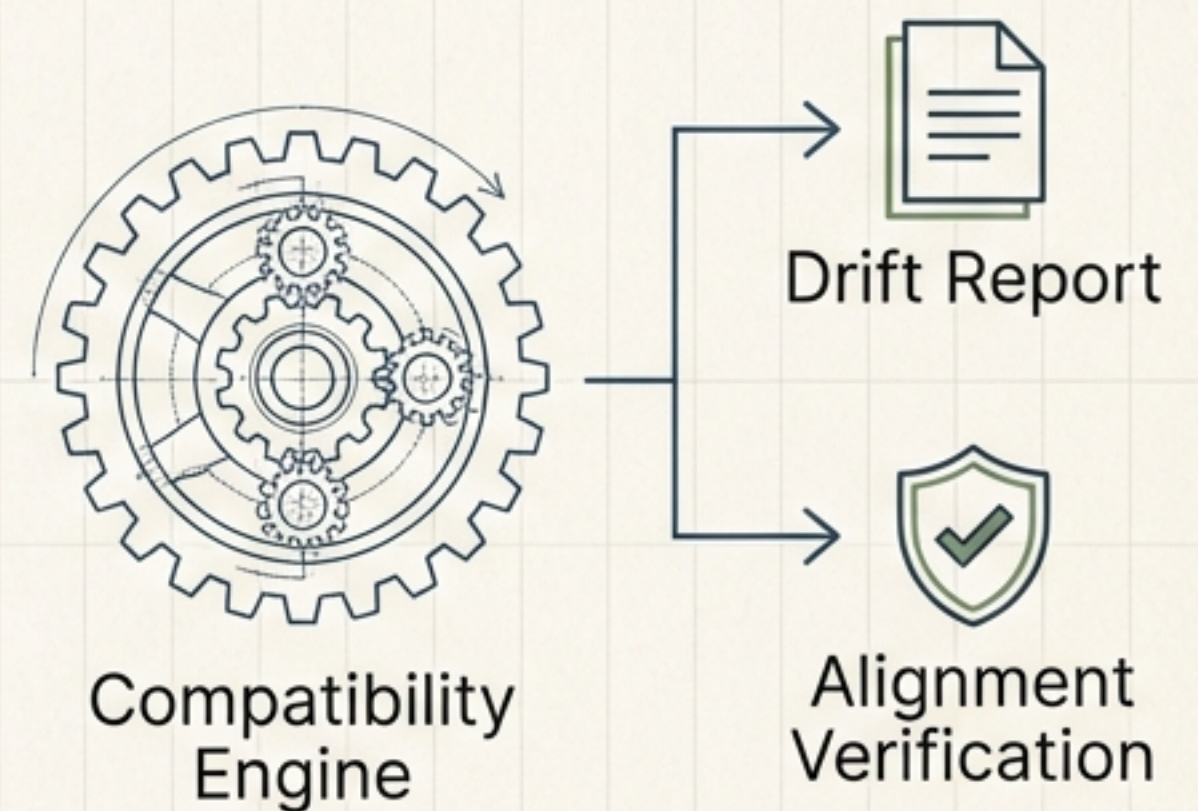
The Engine



Representation

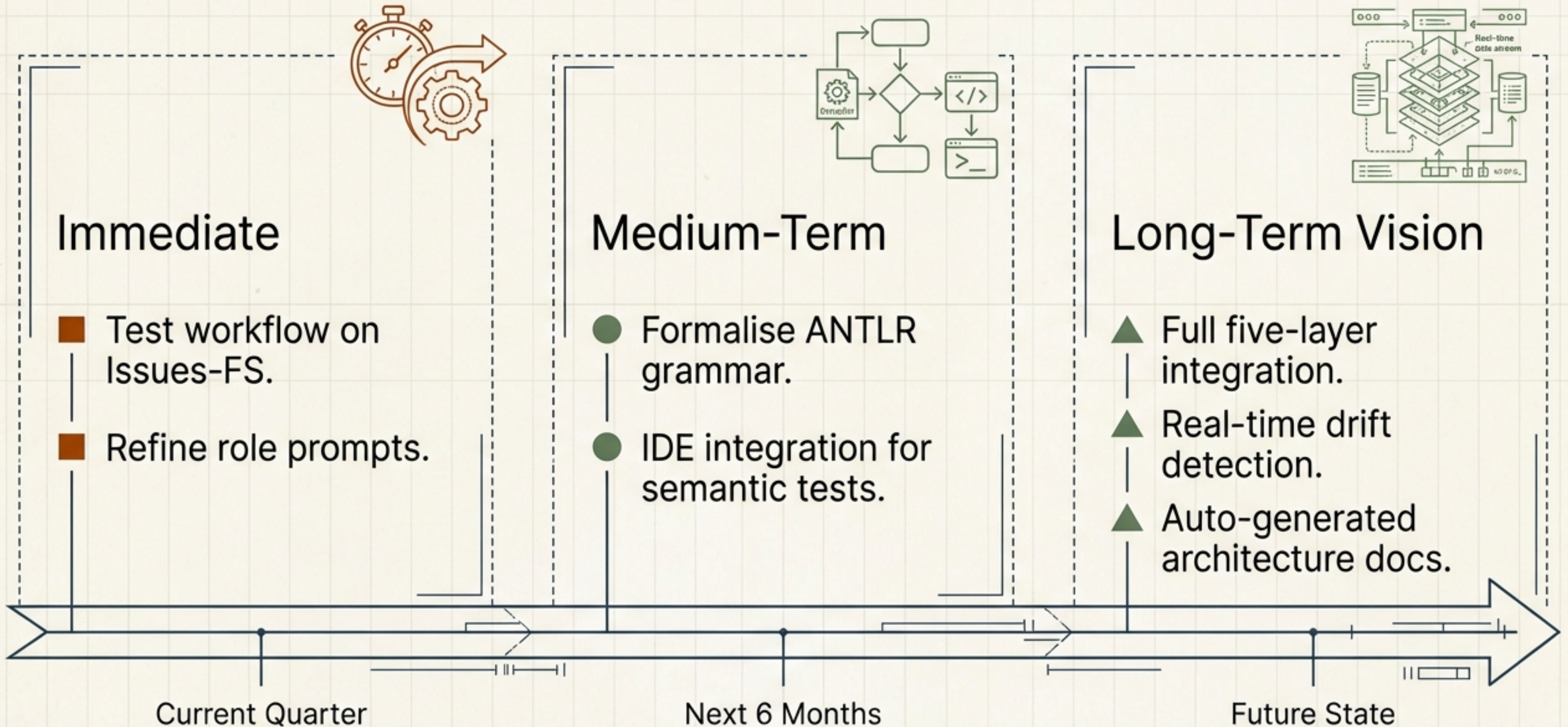


Verification

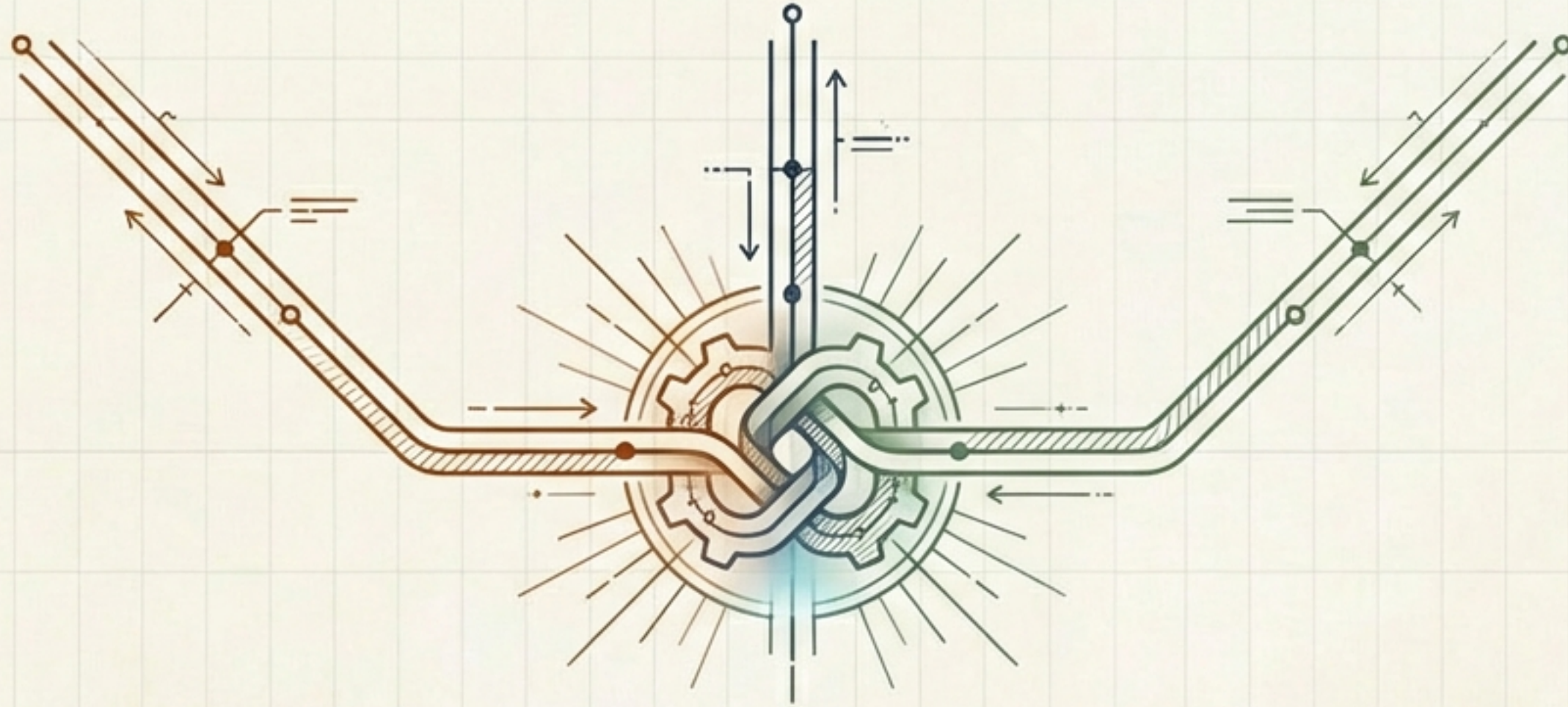


Text as Graph	✓
O&T First	✓
Graph Diff	✓
Fractal Levels	✓

Roadmap to Self-Documenting Systems



Meaning Requires Agreement



A word has meaning when it connects to a definition.
A system has meaning when all its representations tell the same story.

Build for Compatibility.

